# Can Team Composition Predict Victory?


## Abstract

In a team-based game, choosing the right combination of characters should matter. I tested whether the identities of six brawlers could predict the winning team in my own Brawl Stars match history. For that, I manually assembled 162 independent 3v3 matches (121 wins, 41 losses), implemented binary logistic regression and gradient descent, and compared six experimental conditions. Across five stratified folds, the simple historical-win-rate baseline achieved the best overall log loss (0.566); the strongest composition model scored 0.576. On 93 held-out Brawl Ball matches, the all-mode signed-composition model achieved the best log loss (0.574), while a Brawl Ball-only model scored 0.626 without mirroring and 0.664 with mirroring. The exercise shows why data provenance, leakage control, and meaningful baselines matter more than apparent training fit.


## Motivation

Brawl Stars is a real-time competitive game in which three-character teams face one another in modes with different objectives. A useful drafting assistant would estimate how likely a lineup is to win against an opposing lineup before the match starts. My initial question was simple: if I provide the six brawler identities, can a model predict whether Blue (the side recorded as mine) or Red will win? A more ambitious aspiration is to isolate composition strength from the ability of the individuals playing those characters.

In my personal match history Blue wins almost three out of four games, so a model can seem accurate by merely predicting my team. I therefore compared composition-based methods with a prior-only baseline, tested the effects of mirroring match sides, explored two feature encodings, and restricted training to Brawl Ball to see whether a more homogeneous subset was easier to learn from. The goal is to understand how much we can achieve from a limited number of samples to see if this matches with our current understanding of Brawl Starts compositions (mostly qualitative).

## Data collection, cleaning, and augmentation

I began with a 25-match pilot drawn from battle-history screenshots. These games were concentrated in one map and strongly overrepresented a few brawlers. I subsequently broadened my character selection and game modes, then intentionally gathered extra Brawl Ball matches to test specialization. The final manually transcribed dataset contains 162 independent matches, 107 distinct observed brawler labels, and six recorded modes: Brawl Ball (93), Gem Grab (30), Knockout (25), Hot Zone (8), Heist (4), and Basket Brawl (2). Outcomes were 121 Blue wins and 41 losses (74.7% Blue wins). These brawler labels are the observed vocabulary, not a claim about the total game roster.

Each record preserves three Blue and three Red brawlers, the binary Blue_Win outcome, game mode, map, and screenshot provenance. Cleaning standardized brawler names, rejected duplicate characters within a team, checked outcome labels, and checked match identifiers and original/mirrored consistency.

No draw outcomes were recorded. Post-match variables such as Result and Trophy_Change, identifiers, screenshot metadata, and Is_Flipped weren't used as model predictors.
To experiment with side symmetry, I generated one synthetic counterpart for every observed match: Blue and Red brawlers were exchanged and Blue_Win was replaced by 1 − Blue_Win. This produced 324 rows, balanced 162/162 by recorded side, but still only 162 independent real outcomes. Synthetic observations were added only to training folds after the original matches had been split. This avoids placing a match in training and its reversed duplicate in validation. Mirroring changes the represented side prior but does not remove the effects of personal skill, teammates, matchmaking, or brawler-selection behavior. Therefore, we should treat mirroring as an attempt to approximate the real strength composition.

# Pipeline and mathematical method

Figure 1 traces the actual sequence used in the notebooks. Five-fold cross-validation was stratified by observed outcome and whether the mode was Brawl Ball. The same 162 real matches were held out exactly once for E0–E3, and the same 93 Brawl Ball originals were held out for the Brawl Ball comparison. Fold sizes were 34, 33, 33, 31, and 31 real test matches. Brawler vocabularies were constructed from each training fold only; an unseen brawler contributes zero for its unavailable feature. Character slot order has no special meaning in our encodings.

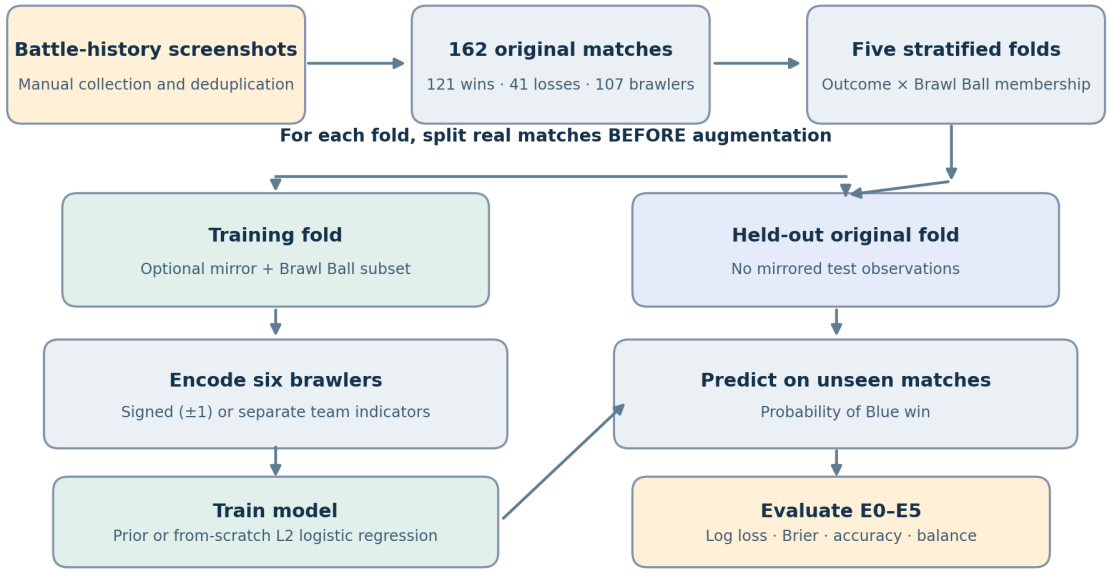



*Figure 1. Leakage-safe, five-fold Brawl Stars prediction pipeline; side-swapped observations are generated exclusively within training folds.*

## Composition features

The main signed representation assigns one feature to each brawler j: x□ = 1(j on Blue) − 1(j on Red). Thus a character exclusive to Blue contributes +1, exclusive to Red −1, and one appearing on both teams contributes 0. A second encoding instead uses separate Blue and Red presence indicators (up to twice as many features). Both approaches treat the three slots within a team as interchangeable; neither explicitly models pairwise synergy, players, or map interactions.

### Logistic regression

For features x and label $y \in \{0,1\}$, logistic regression estimates $p = P(y = 1 \mid x) = \sigma(w^T x + b)$, with $\sigma(z) = 1 / (1 + e^{-z})$. I implemented stable sigmoid evaluation, binary cross-entropy, and batch gradient descent using Python rather than calling a model-fitting library. The optimized objective was:

$$L(w,b) = -(1/N) \Sigma_i [\, y_i \ln p_i + (1-y_i) \ln(1-p_i) \,] + (\lambda/2) \Sigma_j w_j^2$$

After accumulating the gradients over the training examples, parameters were updated according to $w_j \leftarrow w_j - \alpha[(1/N)\Sigma_i(p_i-y_i)x_{ij} + \lambda w_j]$ and $b \leftarrow b - \alpha(1/N)\Sigma_i(p_i-y_i)$. The notebook uses 600 epochs, learning rate $\alpha = 0.3$, and L2 strength $\lambda = 0.01$ for each fitted logistic model. These settings were fixed for the comparisons instead of comparing each method's optimized hyperparameters. The baseline E0 requires no optimization: it predicts the win fraction of the training originals.

## Experimental design

The six conditions change one aspect of the pipeline at a time where possible. E0 is the training-prior baseline; E1 fits signed brawler features on original all-mode matches; E2 uses the same features with mirrored training rows; E3 uses separate-team features on original all-mode matches; E4 trains signed features exclusively on mirrored Brawl Ball data; and E5 trains signed features on original Brawl Ball matches only. Comparing E1 with E0 tests added composition information; E1 with E2 tests augmentation; E1 with E3 tests representation; E1 with E5 and E2 with E4 test specialization while holding mirroring fixed. E4 versus E5 isolates mirroring within Brawl Ball.
Predictions on held-out real matches were aggregated across the five folds. Accuracy uses a 0.5 classification threshold. Balanced accuracy averages the recalls for wins and losses; log loss and Brier score measure quality of predicted probabilities. Log loss is the primary metric because the useful eventual product is a win probability, not just a binary verdict. No synthetic held-out observations were counted as new independent evidence.

## Results

### Overall prediction: 162 held-out original matches

| Model | Acc. | Bal. acc. | Log loss | Brier |
|---|---|---|---|---|
| E0 Prior only | 0.747 | 0.500 | 0.566 | 0.189 |
| E1 Signed, original | 0.722 | 0.516 | 0.576 | 0.192 |
| E2 Signed, mirrored | 0.611 | 0.554 | 0.638 | 0.224 |
| E3 Separate teams, original | 0.747 | 0.532 | 0.584 | 0.196 |

*Table 1. Comparable out-of-fold results on all 162 real matches (E0–E3). Larger accuracy is better; smaller log loss and Brier scores are better.*

The prior-only model produced the lowest overall log loss (0.566) and Brier score (0.189), despite ignoring every brawler. E1 was slightly worse on log loss (0.576) and accuracy (72.2% vs. 74.7%), although balanced accuracy rose marginally to 0.516. E3 retained 74.7% accuracy and modestly improved balanced accuracy (0.532) but worsened log loss to 0.584. E2's mirroring reduced accuracy to 61.1% and

worsened log loss to 0.638, while lifting balanced accuracy to 0.554. None of these results establishes generalizable draft strength beyond the dominant player-side prior.

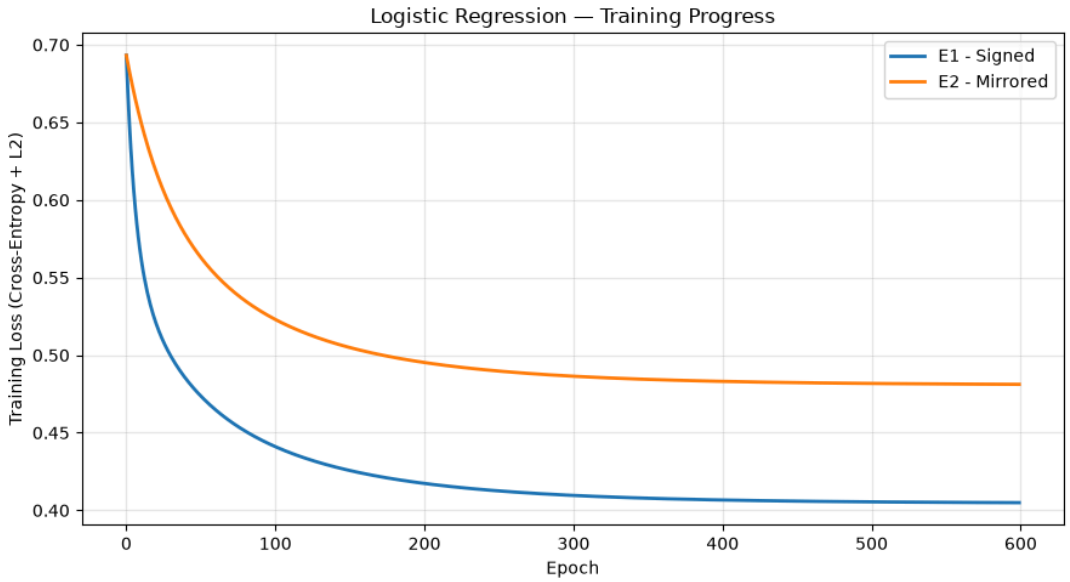



*Figure 2. First-fold training progress for signed original and mirrored logistic regression, showing regularized training loss across 600 epochs.*

Both training-loss curves decline and flatten (approximately 0.405 for E1 and 0.481 for E2 by epoch 600 in the illustrated fold), showing that optimization proceeded. Lower training loss is not evidence of better out-of-sample predictions. In this setting, both composition-based approaches fit the training data substantially more closely than their held-out performance would justify.

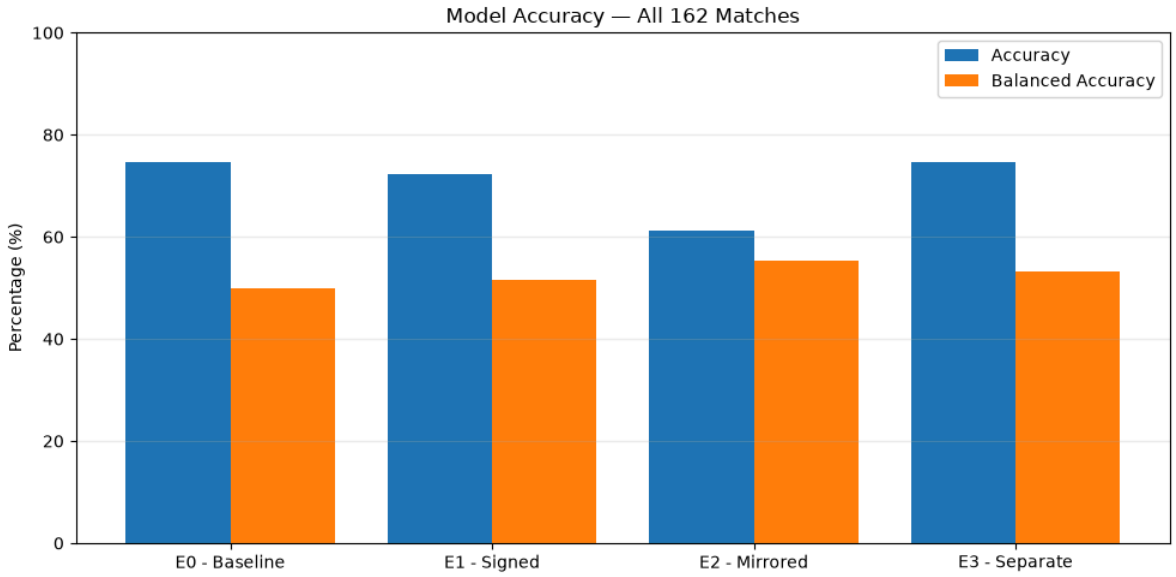



*Figure 3. Accuracy and balanced accuracy across E0–E3, evaluated on all 162 original matches.*

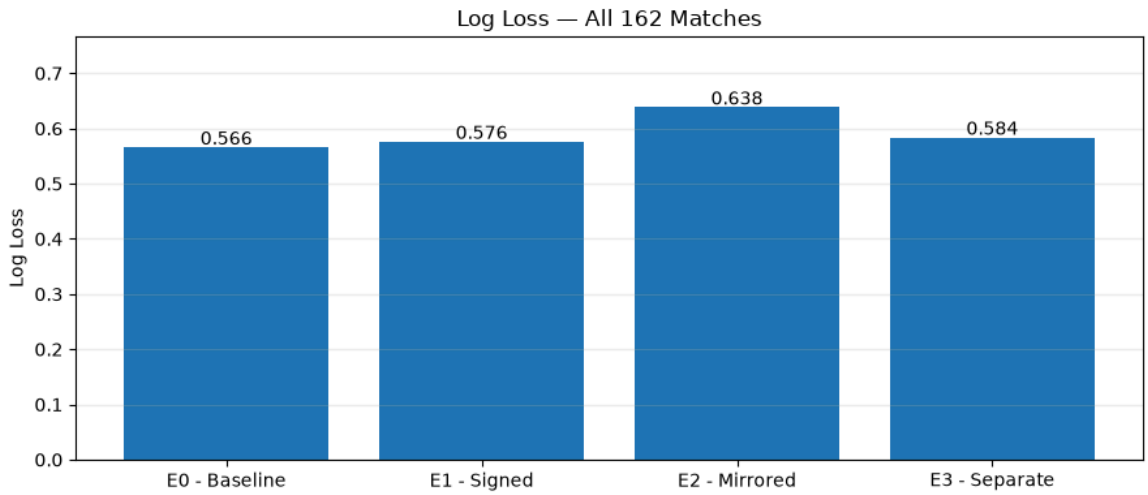



*Figure 4. Overall held-out log loss for E0–E3. Lower is better; the simple training-prior baseline performs best.*

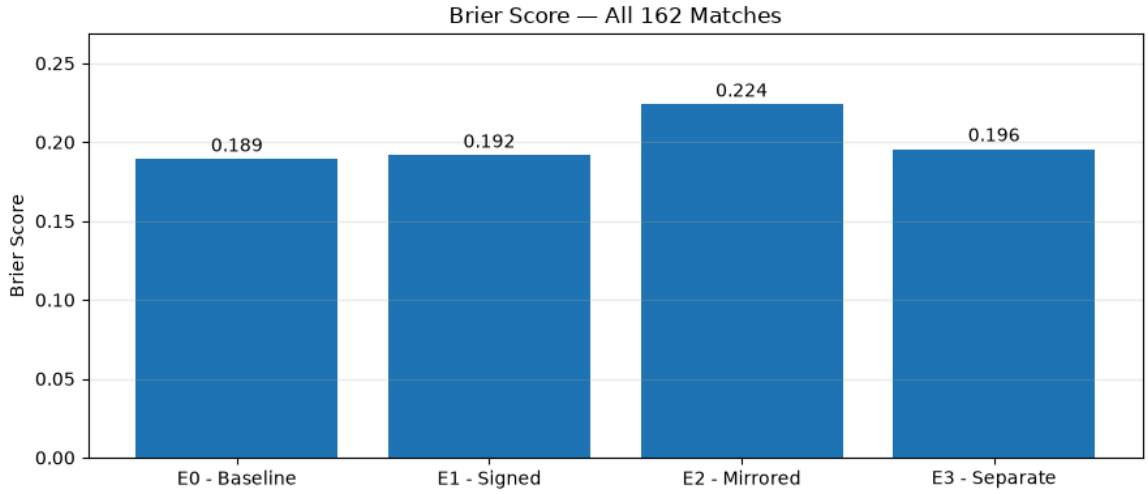



*Figure 5. Overall held-out Brier scores for E0–E3, reinforcing the log-loss comparison.*

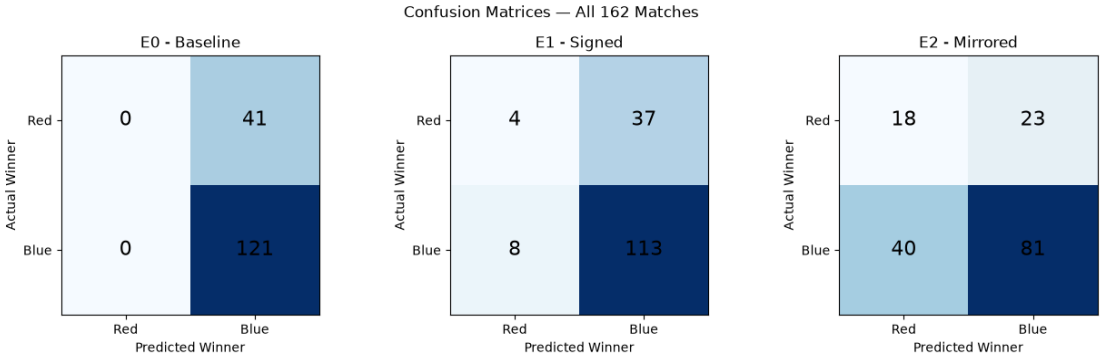



*Figure 6. Confusion matrices on 162 real matches: E0 predicts Blue every time; E1 recognizes 4 of 41 Red wins; E2 recognizes 18 of 41 but also misclassifies 40 actual Blue wins.*

The confusion matrices clarify the trade-off. E0 predicts Blue on every case, correctly classifying 121 wins but missing all 41 losses. E1 correctly predicts four losses and incurs eight missed Blue wins; E2 correctly predicts 18 losses but incorrectly calls 40 Blue victories. Thus E2's higher balanced accuracy

should not be mistaken for superior overall calibration or accuracy on the observed original-match population.

### Brawl Ball specialization: 93 identical held-out matches

| Model | Acc. | Bal. acc. | Log loss | Brier |
|---|---|---|---|---|
| E0 Prior only | 0.731 | 0.500 | 0.583 | 0.197 |
| E1 Signed, original | 0.699 | 0.516 | 0.574 | 0.194 |
| E2 Signed, mirrored | 0.645 | 0.618 | 0.613 | 0.214 |
| E3 Separate teams, original | 0.731 | 0.538 | 0.613 | 0.206 |
| E4 Brawl Ball, mirrored | 0.613 | 0.571 | 0.664 | 0.236 |
| E5 Brawl Ball, original | 0.688 | 0.521 | 0.626 | 0.214 |

*Table 2. All six models evaluated on the SAME 93 held-out original Brawl Ball matches. The Brawl Ball models were trained only on Brawl Ball training observations.*

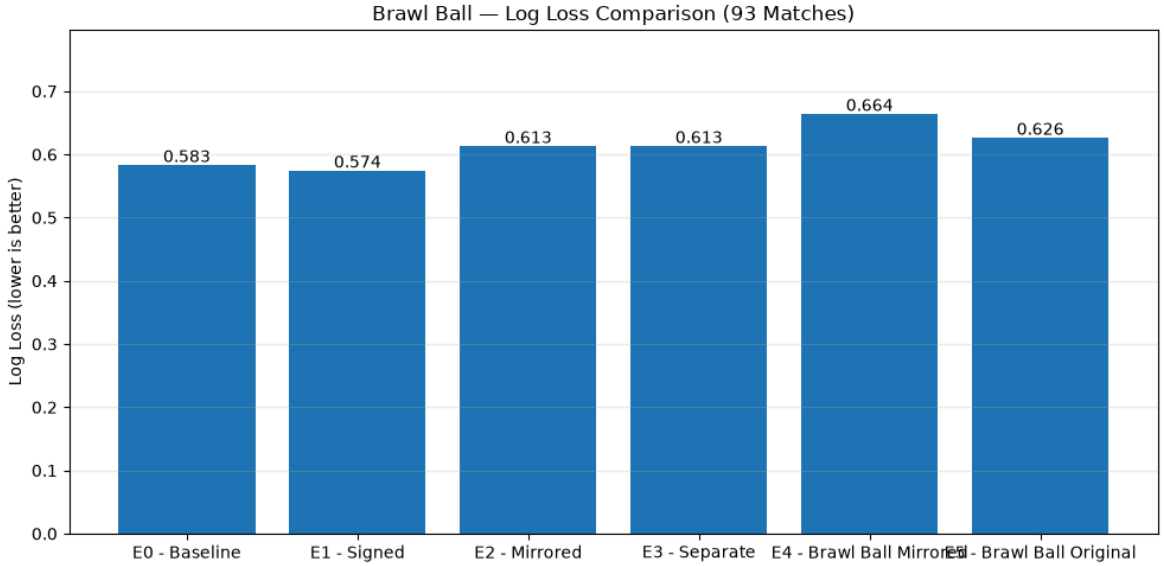



*Figure 7. Log-loss comparison on 93 held-out Brawl Ball matches. E1 is best despite having been trained across game modes.*

The Brawl Ball-only original model E5 yielded log loss 0.626 versus E1's 0.574 on the same test matches: restricting the mode worsened log loss by 0.052. With mirroring, E4 scored 0.664 versus E2's 0.613, a further 0.051 deterioration. Within Brawl Ball-only training, removing mirroring improved log loss from 0.664 (E4) to 0.626 (E5). The most successful Brawl Ball probability estimates actually came from the general, unmirrored signed model E1. Reducing variation in match modes may have removed heterogeneity, but it also discarded valuable examples; the number of potential brawler coefficients remained large relative to the subset.

## Discussion and limitations

The strongest explanation for the baseline's success is the sampling population: these are my matches, not a random cross-section of equally skilled teams. Blue represents my side in the originals, and it wins 74.7% of the time. A predictor that disregards compositions already has a strong prior. The composition features appear not to improve log loss sufficiently to offset the variance from estimating many brawler effects in a small sample. This is consistent with data sparsity and possible overfitting, but a

learning-curve experiment comparing both training and validation loss would be needed to diagnose overfitting more directly.

Mirroring helped a different objective: recognizing more actual losses, reflected in improved balanced accuracy. But the synthetically balanced side distribution does not match the 74.7% Blue-win frequency of the observed originals. Consequently, a model trained to treat Blue and Red symmetrically can assign probabilities that are less suited to predicting outcomes in this specific player-centered archive. This distinction is crucial: more balanced labels do not automatically mean a less confounded dataset, and evaluating on the original real matches is necessary to reveal the trade-off.

Specializing in Brawl Ball likewise did not improve log loss in the controlled tests. E5 and E4 use fewer independent matches than their all-mode counterparts; the subset may still vary by map, teammates, personal character choice, and game balance. Moreover, our feature representation adds individual character contributions linearly; it does not capture synergy, counters, player-brawler skill, or patch-specific interactions. These experiments are cross-validation comparisons on just 162 independent observations (93 Brawl Ball), without confidence intervals or a separate prospective test set; small metric differences should not be treated as conclusive evidence of a population-level improvement.

## Conclusion and next steps

A working, leakage-aware binary classification pipeline can be built entirely from manually collected personal game history and first-principles logistic regression. Yet the clearest finding was negative: team composition did not consistently beat an already strong win-rate baseline on this dataset. Mirroring traded probability quality for somewhat better loss detection, while Brawl Ball restriction reduced rather than improved held-out probability quality. More independent games from diverse players, explicit map and balance-patch metadata, and tests of pairwise character interactions would be more informative next steps than adding complexity to the present small sample. The project demonstrates that a meaningful ML result can be a well-controlled account of where apparent predictive signal does not generalize.

# AI Statement

I used AI in this assignment to (1) register the Brawl Stars matches in a structured format into the Google Sheets*, (2) help me with the setup of the GitHub (sub)repository for this assignment (in the assignments folder), (3) write code according to previous examples used in breakouts in class, in PCWs, along with some of my personal projects, as well as to (4) help me draft the final report after having the working code. My ChatGPT chat can be accessed here and here.

*Interestingly enough, it was so bad at identifying the characters by their photos that I ended up having to do the job mostly manually up until the 200th entry, which was when it started to memorize the brawlers correctly.

# Reproducibility and references

Source data: CS156 Brawl Stars Battle Composition Dataset (Google Sheets): https://docs.google.com/spreadsheets/d/1PT7GDSb56HKSjGbvmj5-bGVWCmbP8Wo7-tSkRxEk7aQ/edit

Code and collection history: CS156 Machine Learning / assignments/01_brawl_stars (GitHub): https://github.com/hammondutra/cs156-machine-learning/tree/main/assignments/01_brawl_stars

Implementation files used for this analysis: data_setup.py, regression_implementation.ipynb, visualize_results.ipynb (provided alongside this report). The experimental notebooks use Python's standard library for data handling and model fitting, and Matplotlib to visualize results. The plots and numbers above are taken from their executed outputs.